

Internship Report: AWS in a College Lost & Found Portal

Introduction

Today's internship session was about Amazon Web Services (AWS). Instead of writing only the definitions of AWS services, I tried to understand how they can be used in a practical project. I selected a College Lost & Found Portal because every college faces the problem of students losing items such as ID cards, wallets, calculators, earphones, water bottles, and notebooks. AWS can make such a portal fast, secure, and easy to manage.

Why Cloud Computing?

Normally, if a college creates this portal on its own computer, it has to maintain the hardware, electricity, storage, backups, and security. By using cloud computing, these responsibilities are handled by AWS. The college can increase or decrease resources whenever required and only pays for what it actually uses. This makes the solution cost-effective, especially for educational institutes.

Amazon EC2

Amazon EC2 can be used to host the main web application. Students, faculty members, and administrators open the website through this server. During college festivals or admission periods, more students may access the portal. In that case, the EC2 instance can be upgraded without changing the application. This provides flexibility and better performance.

AWS Lambda

Lambda is useful for background tasks. Whenever a student reports a lost item, Lambda can automatically send an email to the administrator. If another student uploads a matching found item, Lambda can also notify the owner instantly. Since Lambda runs only when an event occurs, there is no need to keep a server running all day, which helps reduce cost.

Amazon S3

Every lost or found item usually includes one or more photographs. These images can be stored in Amazon S3. Keeping images in S3 instead of the application server improves storage management and makes file access easier. Old images can also be moved to cheaper storage classes if they are no longer required.

Database Services

Important information such as student name, enrollment number, item description, date, location, and claim status can be stored in Amazon RDS. If the application needs very fast access to frequently searched data, services like DynamoDB can also be considered. Using managed databases reduces maintenance work and improves reliability.

IAM and Security

AWS Identity and Access Management (IAM) ensures that every user has only the required permissions. Students can create and view their own reports, while administrators can verify ownership, remove fake reports, and manage users. Proper access control helps protect personal information and prevents misuse of the system.

Load Balancer and Reliability

If many students use the portal at the same time, a Load Balancer can distribute requests across multiple EC2 servers. This reduces the chances of the website becoming slow or unavailable. Regular backups and monitoring also improve reliability.

Conclusion

From today's internship session, I understood that AWS is not only useful for large companies but also for simple college applications. Services like EC2, Lambda, S3, RDS, and IAM can work together to build a secure and scalable Lost & Found Portal. This example helped me understand how different AWS services solve real-world problems instead of only learning their definitions.